

CO4 AT 1-CODING CHALLENGE

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1. Develop a Dynamic Programming algorithm to find the minimum cost path in a given cost matrix. A traveler starts from the top-left corner of the grid and must reach the bottom-right corner. At each step, the traveler is allowed to move either to the right, downward, or diagonally down-right. Each cell contains a specific cost, and the objective is to determine the path with the minimum total cost

CODE:

```
1  def minCost(grid):
13     # First column
14     for i in range(1, n):
15         dp[i][0] = dp[i-1][0] + grid[i][0]
16
17     # Fill remaining cells
18     for i in range(1, n):
19         for j in range(1, m):
20             dp[i][j] = grid[i][j] + min(
21                 dp[i-1][j],
22                 dp[i][j-1],
23                 dp[i-1][j-1]
24             )
25
26     return dp[n-1][m-1]
27
28
29 grid = [
30     [1, 2, 3],
31     [4, 8, 2],
32     [1, 5, 3]
33 ]
```

OUTPUT:

Output

```
Minimum Cost = 8
```

```
=== Code Execution Successful ===
```